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Graded Assignment • 2h

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English Due Jan 5, 11:59 PM IST

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Module 7 Introduction

Lesson 1: Network Pruning

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Module 7 Summative Assessment

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Module 7 Summary

Module 7 Summative Assessment

Review Learning Objectives

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Assignment details

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Instructions

1. What does "pruning" in network compression refer to? **1 / 1 point**

- Adding additional weights to the network
- Expanding the network's layers
- Removing redundant weights or neurons from the network
- Increasing the number of parameters

Correct
Correct!

2. What does the term "Dynamic Computation" refer to in the context of neural networks? **1 / 1 point**

- A network's ability to perform varying amounts of computation depending on the input.
- The dynamic nature of backpropagation during training.
- Adjusting computational resources based on network load.

Correct
Correct!.

3. What does "runtime neural pruning" aim to achieve during inference? **1 / 1 point**

- To increase the inference time for more accuracy.
- To adapt the network's structure for each individual input dynamically.
- To prune the network once before deployment.
- To enhance the training speed by pruning neurons.

Correct
Correct!

4. What is the main benefit of Dynamic Depth in neural network architectures? **0 / 1 point**

- It allows for deeper networks with no impact on computation.
- It dynamically adjusts the number of layers during training for efficiency.
- It increases the network's depth on-the-fly during inference.
- It decreases the need for pre-training the network.

Incorrect
Incorrect. Dynamic Depth adjustments typically happen during training, not inference.

5. Why is it important not to prune too much at once from a neural network? **1 / 1 point**

- To avoid overfitting
- To maintain the structural integrity and allow for recovery
- To increase the training time
- To reduce computational costs

Correct
Correct!

6. What is the primary benefit of weight clustering in network compression? **1 / 1 point**

- To improve the training time of the network.
- To cluster similar weights, thereby reducing the model size.
- To increase the number of parameters for better accuracy.
- To expand the size of the network.

Correct
Correct!

7. Which of the following best describes the Lottery Ticket Hypothesis in network pruning? **1 / 1 point**

- The best network architecture can be won in a lottery.
- A randomly initialized network is as good as a trained one.
- A sub-network with a winning initialization can match the performance of the original network.
- Pruning is a randomized process without specific criteria.

Correct
Correct!

8. What is the concept of Dynamic Width in network design? **1 / 1 point**

- Adjusting the width of the network based on the complexity of the task at hand.
- Increasing the width of the network to improve accuracy unconditionally.
- Statically setting the network width at the design stage.
- Reducing the width of the network during the backpropagation phase.

Correct
Correct!

9. What is the role of "Knowledge Distillation" in network compression? **1 / 1 point**

- Transferring knowledge from a larger, teacher network to a smaller, student network.
- Extracting knowledge exclusively from data without a teacher network.
- Increasing the knowledge capacity of a network by expanding its size.
- Compressing the data before feeding it to the network.

Correct
Correct!

10. How does "Depthwise Separable Convolution" contribute to network efficiency? **1 / 1 point**

- By combining depthwise and pointwise convolutions to reduce parameters.
- By increasing the depth of convolutions for better feature extraction.
- By separating the convolution operation into individual channels.
- By using depthwise convolutions to deepen the network without cost.

Correct
Correct!

11. What is the advantage of using depthwise separable convolution compared to standard convolution? **1 / 1 point**

- It requires more parameters to achieve higher accuracy.
- It decreases model size and computational requirements by separating filtering and combining steps.
- It allows for deeper network architectures.
- It increases the interaction between channels.

Correct
Correct!

12. Which technique is used in network compression to maintain high performance with a reduced model size? **1 / 1 point**

- Increasing the size of the training dataset.
- Expanding the number of layers in the network.
- Implementing deeper convolutional layers.
- Applying techniques such as pruning, quantization, and knowledge distillation.

Correct
Correct!

13. When discussing "Architecture Design" for network compression, what is typically a primary consideration? **1 / 1 point**

- The aesthetic appearance of the network's graphical representation.
- The ability of the network to self-adjust its architecture during training.
- The physical design of the chips on which the network will run.
- The theoretical maximum depth and width of neural networks.

Correct
Correct!

14. In network pruning, what does fine-tuning refer to? **1 / 1 point**

- The initial training of the network.
- The detailed design of the network architecture.
- Adjusting the learning rate during training.
- Retraining the network after pruning to regain performance.

Correct
Correct!

15. How does knowledge distillation aid in network compression? **1 / 1 point**

- By increasing the number of parameters to enhance knowledge transfer.
- By transferring knowledge from a larger model (teacher) to a smaller one (student), improving the student model's performance.
- By distilling the network into a simpler architecture without training.
- By reducing the knowledge within a network to its most essential components only.

Correct
Correct!